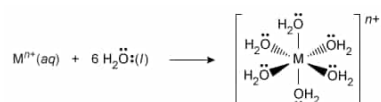


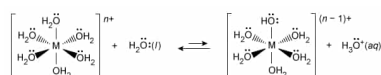
Hydrolysis is a process in which a chemical substance decomposes by reacting with water. In the hydrolysis of certain ionic salts, water itself can be decomposed to some extent via an acid-base reaction in which a cation acts as an acid or an anion acts as a base.

Although metal cations (M^{n+}) alone have no H^+ ions to donate, some cations can become hydrated by forming coordinate bonds with neighboring water molecules in solution, as shown in Reaction 1.



Reaction 1

In the resulting complex ion, the central cation shifts electron density away from the H atoms by an inductive effect, making it possible for the complex to donate H^+ ions to water molecules in the solution and form $H_3O^+(aq)$, as shown in Reaction 2. Consequently, aqueous solutions of small, highly charged metal ions (ie, $M^{n+}(aq)$ where $n \geq 2$) are often weakly acidic.



Reaction 2

To investigate the properties of $Zn^{2+}(aq)$ and $Al^{3+}(aq)$, a chemist performed the following experiments.

Experiment 1

A sample of a zinc salt was dissolved in water to produce a 0.10 M $Zn^{2+}(aq)$ solution. The solution was checked with a pH probe, and the measured pH confirmed that the solution was weakly acidic. $[Zn(H_2O)_6]^{2+}(aq)$ was found to have a pK_a of 9.0, as reported in the scientific literature.

Experiment 2

A sample of an aluminum salt was dissolved in water to produce a 0.10 M $Al^{3+}(aq)$ solution. The solution was observed to turn the blue litmus paper red and had a measured pH of 3.0. The pK_a of $[Al(H_2O)_6]^{3+}(aq)$ was not found in the scientific literature.

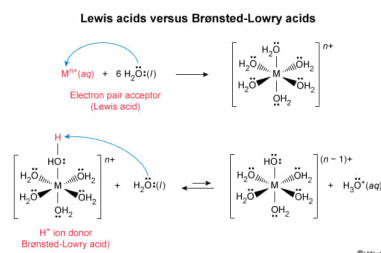
What roles do M^{n+} and $[M(H_2O)_6]^{n+}$ have in Reactions 1 and 2?

- ☐ A. Both $M^{n+}(aq)$ and $[M(H_2O)_6]^{n+}(aq)$ act *only* as Lewis acids. [8%]
- ☒ B. $M^{n+}(aq)$ acts as a Lewis acid, whereas $[M(H_2O)_6]^{n+}(aq)$ acts as a Brønsted-Lowry acid. [75%]
- ☐ C. $M^{n+}(aq)$ acts as a Brønsted-Lowry acid, whereas $[M(H_2O)_6]^{n+}(aq)$ acts as a Lewis acid. [13%]
- ☐ D. Both $M^{n+}(aq)$ and $[M(H_2O)_6]^{n+}(aq)$ act as Brønsted-Lowry acids. [3%]

Correct

75% answered correctly

Explanation:



By the **Brønsted-Lowry definitions**, an **acid** is a species that **donates a proton** (ie, an H^+ ion) and a **base** is a species that **accepts a proton**. Although the Brønsted-Lowry definitions are sufficient to identify acids and bases in many cases, these definitions are inadequate for species that lack acidic protons in their structure yet exhibit acid-base behavior. To overcome this limitation, the **Lewis definitions** were developed. Accordingly, a **Lewis acid** is defined as an **electron pair acceptor**, and a **Lewis base** is defined as an **electron pair donor**.

In Reaction 1, $M^{n+}(aq)$ acts as a **Lewis acid** by accepting an electron pair from each of the six H_2O molecules. In Reaction 2, the $[M(H_2O)_6]^{n+}$ complex acts as a **Brønsted-Lowry acid** by donating an H^+ ion to a H_2O molecule in solution.

(Choice A) $[M(H_2O)_6]^{n+}(aq)$ can act as both a Lewis acid *and* a Brønsted-Lowry acid.

(Choice C) The acid definitions are reversed in this answer choice.

(Choice D) $M^{n+}(aq)$ does not act as a Brønsted-Lowry acid because $M^{n+}(aq)$ cannot donate a proton.

Educational objective:

A Brønsted-Lowry acid is a species that donates an H^+ ion, whereas a Lewis acid is a species that accepts an electron pair. Conversely, a Brønsted-Lowry base is a species that accepts an H^+ ion, whereas a Lewis base is a species that donates an electron pair.

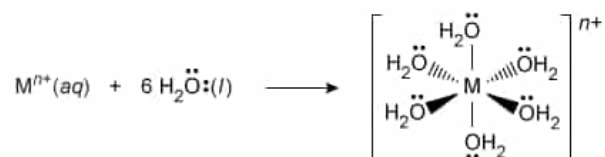
Time spent: 0 sec

Subject:
General Chemistry

QID: 403503

System:
Solutions and Electrochemistry

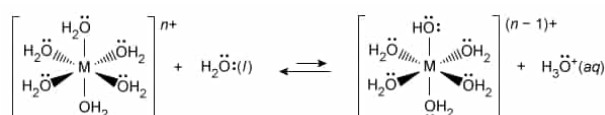
Hydrolysis is a process in which a chemical substance decomposes by reacting with water. In the hydrolysis of certain ionic salts, water itself can be decomposed to some extent via an acid-base reaction in which a cation acts as an acid or an anion acts as a base. Although metal cations (M^{n+}) alone have no H^+ ions to donate, some cations can become hydrated by forming coordinate bonds with neighboring water molecules in solution, as shown in Reaction 1.



Reaction 1

In the resulting complex ion, the central cation shifts electron density away from the H atoms by an inductive effect, making it possible for the complex to donate H^+ ions to water molecules in the solution and form $H_3O^+(aq)$, as shown in Reaction 2.

Consequently, aqueous solutions of small, highly charged metal ions (ie, $M^{n+}(aq)$ where $n \geq 2$) are often weakly acidic.



Reaction 2

To investigate the properties of $Zn^{2+}(aq)$ and $Al^{3+}(aq)$, a chemist performed the following experiments.

Experiment 1

A sample of a zinc salt was dissolved in water to produce a 0.10 M $Zn^{2+}(aq)$ solution. The solution was checked with a pH probe, and the measured pH confirmed that the solution was weakly acidic. $[Zn(H_2O)_6]^{2+}(aq)$ was found to have a pK_a of 9.0, as reported in the scientific literature.

Experiment 2

A sample of an aluminum salt was dissolved in water to produce a 0.10 M $Al^{3+}(aq)$ solution. The solution was observed to turn the blue litmus paper red and had a measured pH of 3.0. The pK_a of $[Al(H_2O)_6]^{3+}(aq)$ was not found in the scientific literature.

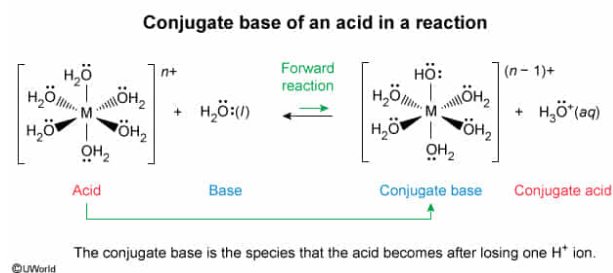
In Reaction 2, which species is the conjugate base in the forward reaction?

- ☐ A. $[\text{M}(\text{H}_2\text{O})_6]^{n+}(\text{aq})$ [3%]
- ☐ B. $\text{H}_2\text{O}(\text{l})$ [11%]
- ✓ ☒ C. $[\text{M}(\text{H}_2\text{O})_5\text{OH}]^{(n-1)+}(\text{aq})$ [75%]
- ☐ D. $\text{H}_3\text{O}^+(\text{aq})$ [8%]

Correct

75% answered correctly

Explanation:



The **conjugate** of an acid (or base) is the species that the acid (or base) becomes after losing (or gaining) one proton (ie, an H^+ ion). A **base** is converted into its **conjugate acid** when the base accepts a proton. Likewise, an **acid** is converted into its **conjugate base** when the acid donates a proton. Therefore, an acid (or base) and its conjugate have chemical formulas that **differ by only one proton**.

In the forward direction of Reaction 2, the $[\text{M}(\text{H}_2\text{O})_6]^{n+}(\text{aq})$ complex acts as an acid by donating a proton to a $\text{H}_2\text{O}(\text{l})$ molecule. Therefore, **$[\text{M}(\text{H}_2\text{O})_5\text{OH}]^{(n-1)+}(\text{aq})$ is the conjugate base** of the acid in the forward reaction (ie, the species that the $[\text{M}(\text{H}_2\text{O})_6]^{n+}(\text{aq})$ acid becomes after losing one proton).

(Choice A) $[\text{M}(\text{H}_2\text{O})_6]^{n+}(\text{aq})$ acts as the *acid* in the forward direction of Reaction 2.

(Choice B) $\text{H}_2\text{O}(\text{l})$ acts as the base in the forward direction of Reaction 2, but this is not the *conjugate base* (ie, the species that the acid becomes after **deprotonation**).

(Choice D) $\text{H}_3\text{O}^+(\text{aq})$ is the *conjugate acid* (ie, the species that the base becomes after being protonated) in the forward direction of Reaction 2.

Educational objective:

The conjugate of an acid (or base) is the species that the acid (or base) becomes

after losing (or gaining) one proton. An acid (or base) and its conjugate have chemical formulas that differ by only one proton.

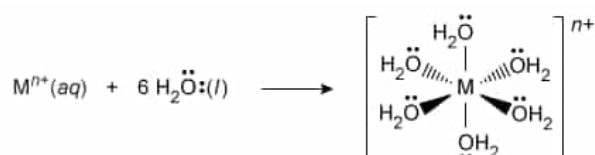
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Subject:
General Chemistry

QID:403504

System:
Solutions and Electrochemistry

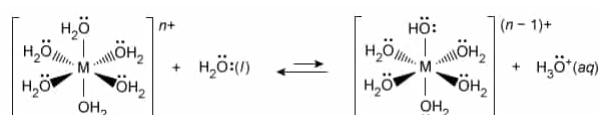
Hydrolysis is a process in which a chemical substance decomposes by reacting with water. In the hydrolysis of certain ionic salts, water itself can be decomposed to some extent via an acid-base reaction in which a cation acts as an acid or an anion acts as a base. Although metal cations (M^{n+}) alone have no H^+ ions to donate, some cations can become hydrated by forming coordinate bonds with neighboring water molecules in solution, as shown in Reaction 1.



Reaction 1

In the resulting complex ion, the central cation shifts electron density away from the H atoms by an inductive effect, making it possible for the complex to donate H^+ ions to water molecules in the solution and form $H_3O^+(aq)$, as shown in Reaction 2.

Consequently, aqueous solutions of small, highly charged metal ions (ie, $M^{n+}(aq)$ where $n \geq 2$) are often weakly acidic.



Reaction 2

To investigate the properties of $Zn^{2+}(aq)$ and $Al^{3+}(aq)$, a chemist performed the following experiments.

Experiment 1

A sample of a zinc salt was dissolved in water to produce a 0.10 M $Zn^{2+}(aq)$ solution. The solution was checked with a pH probe, and the measured pH confirmed that the solution was weakly acidic. $[Zn(H_2O)_6]^{2+}(aq)$ was found to have a pK_a of 9.0, as reported in the scientific literature.

Experiment 2

A sample of an aluminum salt was dissolved in water to produce a 0.10 M $Al^{3+}(aq)$ solution. The solution was observed to turn the blue litmus paper red and had a measured pH of 3.0. The pK_a of $[Al(H_2O)_6]^{3+}(aq)$ was not found in the scientific literature.

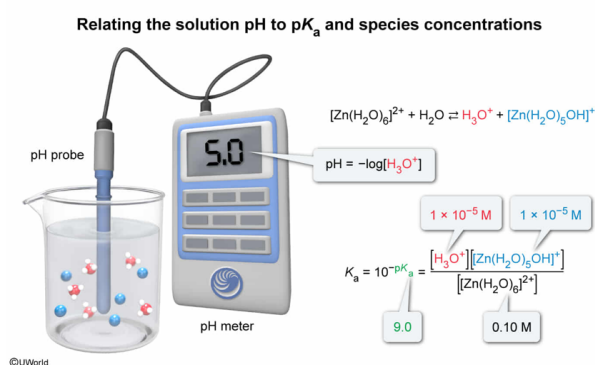
What is the pH of the $\text{Zn}^{2+}(\text{aq})$ solution in Experiment 1?

- ☐ A. 4 [6%]
☒ B. 5 [71%]
☐ C. 7 [8%]
☐ D. 9 [12%]

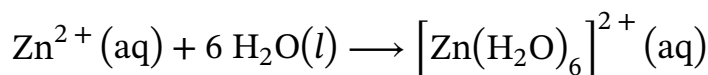
Correct

71% answered correctly

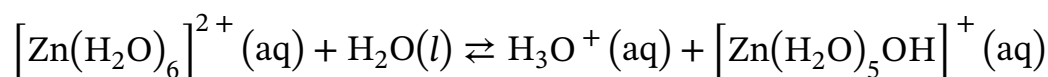
Explanation:



In Experiment 1, the $\text{Zn}^{2+}(\text{aq})$ forms **coordinate bonds** with $\text{H}_2\text{O}(\text{l})$ molecules to produce a hydrated $[\text{Zn}(\text{H}_2\text{O})_6]^{2+}(\text{aq})$ complex in the solution.



The $[\text{Zn}(\text{H}_2\text{O})_6]^{2+}(\text{aq})$ complex acts as a **weak acid**, and a small amount **ionizes into** $\text{H}_3\text{O}^+(\text{aq})$ (ie, a hydronium ion) and $[\text{Zn}(\text{H}_2\text{O})_5\text{OH}]^+(\text{aq})$ (ie, the **conjugate base** of the complex).



Accordingly, the K_a **expression** for the resulting solution equilibrium is:

$$K_a = \frac{[\text{H}_3\text{O}^+][\text{Zn}(\text{H}_2\text{O})_5\text{OH}]^+}{[\text{Zn}(\text{H}_2\text{O})_6]^{2+}}$$

Because the $[\text{Zn}(\text{H}_2\text{O})_6]^{2+}(\text{aq})$ complex forms from $\text{Zn}^{2+}(\text{aq})$ in a 1:1 **mole ratio**, the initial concentration of the complex must be equal to the given initial concentration of the zinc cation:

$$[\text{Zn}^{2+}] = \left[[\text{Zn}(\text{H}_2\text{O})_6]^{2+} \right] = 0.10 \text{ M}$$

During ionization, an equal number of $\text{H}_3\text{O}^+(\text{aq})$ and $[\text{Zn}(\text{H}_2\text{O})_5\text{OH}]^+(\text{aq})$ are produced. As such, their equilibrium concentrations must be equal to the small amount x that ionizes from the initial $[\text{Zn}(\text{H}_2\text{O})_6]^{2+}(\text{aq})$:

$$[\text{H}_3\text{O}^+] = \left[[\text{Zn}(\text{H}_2\text{O})_5\text{OH}]^+ \right] = x$$

The concentrations of the system can be summarized as follows:

	$[\text{Zn}(\text{H}_2\text{O})_6]^{2+}$	$[\text{H}_3\text{O}^+]$	$[\text{Zn}(\text{H}_2\text{O})_5\text{OH}]^+$
Initial:	0.10	0	0
Change:	$-x$	$+x$	$+x$
Equilibrium:	$0.10 - x$	x	x

Substituting the equilibrium concentrations and the $\text{p}K_a$ from the passage into the K_a expression gives:

$$K_a = 10^{-\text{p}K_a} = \frac{[\text{H}_3\text{O}^+][\text{Zn}(\text{H}_2\text{O})_5\text{OH}]^+}{[\text{Zn}(\text{H}_2\text{O})_6]^{2+}}$$

$$10^{-9.0} = \frac{(x)(x)}{0.10 - x}$$

$$1 \times 10^{-9} = \frac{x^2}{0.10 - x}$$

When x is small, the expression can be simplified by approximation as:

$$1 \times 10^{-9} = \frac{x^2}{0.10 - x} \approx \frac{x^2}{0.10}$$

Solving for x gives:

$$x^2 = (1 \times 10^{-9})(0.10) = 1 \times 10^{-10}$$

$$x = [\text{H}_3\text{O}^+] = 1 \times 10^{-5} \text{ M}$$

Therefore, **based on $[\text{H}_3\text{O}^+]$** , the **pH** of the 0.10 M $\text{Zn}^{2+}(\text{aq})$ solution is:

$$\text{pH} = -\log [\text{H}_3\text{O}^+] = -\log (1 \times 10^{-5}) = 5$$

(Choice A) A pH of 4 results from incorrectly evaluating x^2 as 1×10^{-8} during the calculation:

$$x^2 = (1 \times 10^{-9})(0.10) \neq 1 \times 10^{-8}$$

$$x = [\text{H}_3\text{O}^+] \neq 1 \times 10^{-4} \text{ M}$$

(Choice C) A pH of 7 indicates a neutral solution that is neither acidic nor basic, but the passage states that the measured pH confirmed the solution was weakly acidic.

(Choice D) A pH of 9 assumes that $\text{pH} = \text{p}K_a$, but this condition can only happen when the concentrations of the weak acid and its conjugate base are equal, which is not the case for the solution in Experiment 1. In addition, a pH of 9.0 indicates a *basic* solution rather than an acidic solution.

Educational objective:

Small, highly charged metal cations can form hydrated complexes that are weakly acidic. The acid dissociation constant K_a (or $\text{p}K_a$) indicates the extent to which a weak acid ionizes in solution. The pH of the solution is determined by the equilibrium concentration of the hydronium ion produced by the ionization of the acid.

Time spent: 0 sec

Subject:

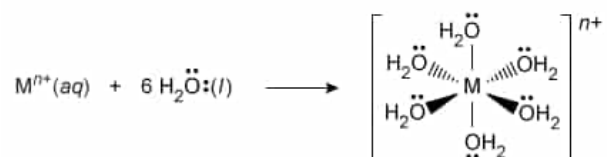
General Chemistry

QID: 403505

System:

Solutions and Electrochemistry

Hydrolysis is a process in which a chemical substance decomposes by reacting with water. In the hydrolysis of certain ionic salts, water itself can be decomposed to some extent via an acid-base reaction in which a cation acts as an acid or an anion acts as a base. Although metal cations (M^{n+}) alone have no H^+ ions to donate, some cations can become hydrated by forming coordinate bonds with neighboring water molecules in solution, as shown in Reaction 1.



Reaction 1

In the resulting complex ion, the central cation shifts electron density away from the H atoms by an inductive effect, making it possible for the complex to donate H^+ ions to water molecules in the solution and form $H_3O^+(aq)$, as shown in Reaction 2.

Consequently, aqueous solutions of small, highly charged metal ions (ie, $M^{n+}(aq)$ where $n \geq 2$) are often weakly acidic.



Reaction 2

To investigate the properties of $Zn^{2+}(aq)$ and $Al^{3+}(aq)$, a chemist performed the following experiments.

Experiment 1

A sample of a zinc salt was dissolved in water to produce a 0.10 M $Zn^{2+}(aq)$ solution. The solution was checked with a pH probe, and the measured pH confirmed that the solution was weakly acidic. $[Zn(H_2O)_6]^{2+}(aq)$ was found to have a pK_a of 9.0, as reported in the scientific literature.

Experiment 2

A sample of an aluminum salt was dissolved in water to produce a 0.10 M $Al^{3+}(aq)$ solution. The solution was observed to turn the blue litmus paper red and had a measured pH of 3.0. The pK_a of $[Al(H_2O)_6]^{3+}(aq)$ was not found in the scientific literature.

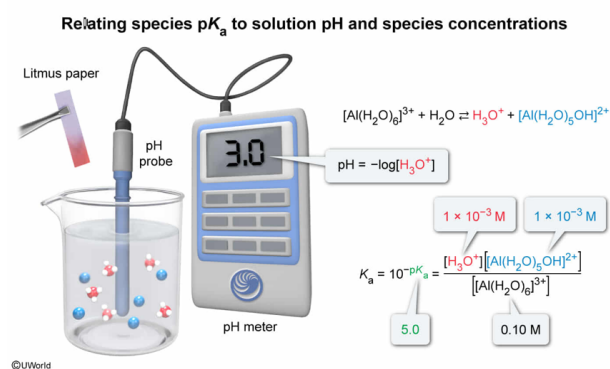
Based on the results from Experiment 2, the pK_a of $[Al(H_2O)_6]^{3+}(aq)$ is:

- ☐ A. 3 [29%]
☒ B. 5 [35%]
☐ C. 7 [15%]
☐ D. 8 [19%]

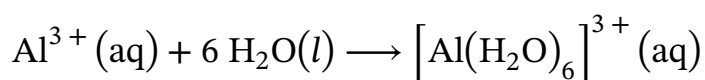
Correct

34% answered correctly

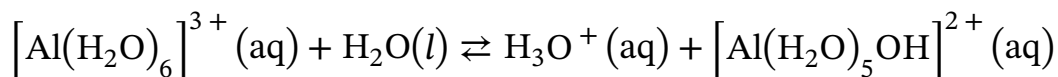
Explanation:



In Experiment 2, the $Al^{3+}(aq)$ forms **coordinate bonds** with $H_2O(l)$ molecules to produce a hydrated $[Al(H_2O)_6]^{3+}(aq)$ complex in the solution.



The $[Al(H_2O)_6]^{3+}(aq)$ complex acts as a **weak acid**, and a small amount **ionizes into $H_3O^+(aq)$** (ie, a hydronium ion) and $[Al(H_2O)_5OH]^{2+}(aq)$ (ie, the **conjugate base** of the complex).



Accordingly, the **K_a expression** for the resulting solution equilibrium is:

$$K_a = \frac{[H_3O^+][Al(H_2O)_5OH]^{2+}}{[Al(H_2O)_6]^{3+}}$$

The passage states that the solution was observed to **turn blue litmus paper red** and had a measured pH of 3.0. The **$[H_3O^+]$** of a solution is **related to** its **pH** according to the relationship:

$$\text{pH} = -\log [\text{H}_3\text{O}^+]$$

Solving for $[\text{H}_3\text{O}^+]$ and substituting in the measured pH value gives:

$$[\text{H}_3\text{O}^+] = 10^{-\text{pH}} = 10^{-3.0} = 1 \times 10^{-3} \text{ M}$$

Because $\text{H}_3\text{O}^+(\text{aq})$ and $[\text{Al}(\text{H}_2\text{O})_5\text{OH}]^{2+}(\text{aq})$ form in a 1:1 **mole ratio**, the concentrations of these species must be equal:

$$[\text{H}_3\text{O}^+] = \left[[\text{Al}(\text{H}_2\text{O})_5\text{OH}]^{2+} \right] = 1 \times 10^{-3} \text{ M}$$

Likewise, $[\text{Al}(\text{H}_2\text{O})_6]^{3+}(\text{aq})$ forms from $0.10 \text{ M Al}^{3+}(\text{aq})$ in a 1:1 mole ratio. As a result, the initial concentrations of these species must be equal:

$$[\text{Al}^{3+}] = \left[[\text{Al}(\text{H}_2\text{O})_6]^{3+} \right]_{\text{initial}} = 0.10 \text{ M}$$

However, at equilibrium the concentration of the complex must decrease by the molar amount that ionizes, which is equivalent to $[\text{H}_3\text{O}^+]$ and $[[\text{Al}(\text{H}_2\text{O})_5\text{OH}]^{2+}]$.

$$\left[[\text{Al}(\text{H}_2\text{O})_6]^{3+} \right]_{\text{equilibrium}} = 0.10 \text{ M} - 1 \times 10^{-3} \text{ M}$$

The concentrations of the system can be summarized as follows:

	$[[\text{Al}(\text{H}_2\text{O})_6]^{3+}]$	$[\text{H}_3\text{O}^+]$	$[[\text{Al}(\text{H}_2\text{O})_5\text{OH}]^{2+}]$
Initial:	0.10	0	0
Change:	-0.0010	+0.0010	+0.0010
Equilibrium:	$0.10 - 0.0010$	0.0010	0.0010

Substituting the equilibrium concentrations into the K_a expression gives:

$$K_a = \frac{[\text{H}_3\text{O}^+] \left[[\text{Al}(\text{H}_2\text{O})_5\text{OH}]^{2+} \right]}{\left[[\text{Al}(\text{H}_2\text{O})_6]^{3+} \right]} = \frac{(1 \times 10^{-3})(1 \times 10^{-3})}{(0.10 - 1 \times 10^{-3})}$$

Because the amount that ionizes is small, the expression can be simplified by approximation as:

$$K_a = \frac{(1 \times 10^{-3})(1 \times 10^{-3})}{(0.10 - 1 \times 10^{-3})} = \frac{(1 \times 10^{-3})(1 \times 10^{-3})}{(0.099)} \approx \frac{(1 \times 10^{-3})(1 \times 10^{-3})}{0.10}$$

Solving for K_a gives:

$$K_a \approx \frac{(1 \times 10^{-3})(1 \times 10^{-3})}{0.10} = \frac{1 \times 10^{-6}}{0.10} = \frac{1 \times 10^{-6}}{1 \times 10^{-1}} = 1 \times 10^{[-6 - (-1)]} = 1 \times 10^{-5}$$

Therefore, the pK_a of $[Al(H_2O)_6]^{3+}(aq)$ is:

$$pK_a = -\log K_a = -\log (1 \times 10^{-5}) = 5$$

(Choice A) A pK_a of 3 incorrectly assumes that $pH = pK_a$, but this condition can only happen when the concentrations of the weak acid and its conjugate base are equal, which is not the case for the solution in Experiment 2.

(Choice C) A pK_a of 7 results from incorrectly evaluating K_a as 1×10^{-7} during the calculation:

$$K_a \approx \frac{(1 \times 10^{-3})(1 \times 10^{-3})}{0.10} = \frac{1 \times 10^{-6}}{1 \times 10^{-1}} \neq 1 \times 10^{-7}$$

(Choice D) A pK_a of 8 results from incorrectly evaluating K_a as 1×10^{-8} during the calculation:

$$K_a \approx \frac{(1 \times 10^{-3})(1 \times 10^{-3})}{0.10} \neq \frac{1 \times 10^{-9}}{1 \times 10^{-1}} = 1 \times 10^{-8}$$

Educational objective:

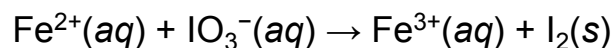
Small, highly charged metal cations can form hydrated complexes that are weakly acidic. The acid dissociation constant K_a (or pK_a) indicates the extent to which a weak acid ionizes in solution. The pH of the solution is determined by the equilibrium concentration of the hydronium ion produced by the ionization of the acid.

Time spent: 0 sec

Subject:
General Chemistry

QID: 403506

System:
Solutions and Electrochemistry



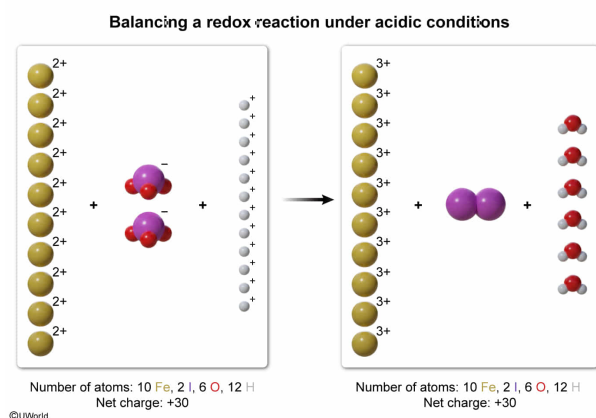
Consider the above unbalanced redox reaction occurring under acidic conditions. Which of the following correctly represents the balanced reaction?

- ☐ A. $\text{Fe}^{2+}(\text{aq}) + 2 \text{IO}_3^{-}(\text{aq}) + 12 \text{H}^{+}(\text{aq}) \rightarrow \text{Fe}^{3+}(\text{aq}) + \text{I}_2(\text{s}) + 6 \text{H}_2\text{O}(\text{l})$ [29%]
- ☐ B. $5 \text{Fe}^{2+}(\text{aq}) + \text{IO}_3^{-}(\text{aq}) + 6 \text{H}^{+}(\text{aq}) \rightarrow 5 \text{Fe}^{3+}(\text{aq}) + \text{I}_2(\text{s}) + 3 \text{H}_2\text{O}(\text{l})$ [15%]
- ✓ ☒ C. $10 \text{Fe}^{2+}(\text{aq}) + 2 \text{IO}_3^{-}(\text{aq}) + 12 \text{H}^{+}(\text{aq}) \rightarrow 10 \text{Fe}^{3+}(\text{aq}) + \text{I}_2(\text{s}) + 6 \text{H}_2\text{O}(\text{l})$ [49%]
- ☐ D. $10 \text{Fe}^{2+}(\text{aq}) + 2 \text{IO}_3^{-}(\text{aq}) + 6 \text{H}_2\text{O}(\text{l}) \rightarrow 10 \text{Fe}^{3+}(\text{aq}) + \text{I}_2(\text{s}) + 12 \text{OH}^{-}(\text{aq})$ [5%]

Correct

49% answered correctly

Explanation:

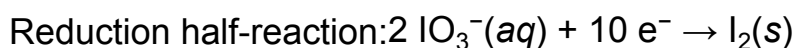


An **oxidation-reduction (redox) reaction** is one in which a reactant is **oxidized** while another reactant is simultaneously **reduced**. As such, the **overall redox reaction** is the **sum of** the oxidation and reduction **half-reactions**. In a **balanced redox reaction**, the total number of **electrons released** from the oxidation half-reaction **must be equal** to the number of **electrons used** by the reduction half-reaction. The first step in balancing a redox reaction is to divide the overall reaction into its respective half-reactions and balance the electrons exchanged.

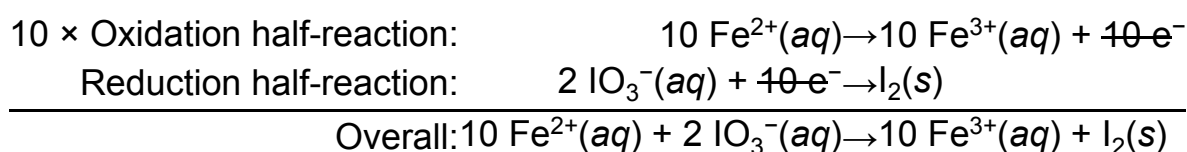
In the provided redox reaction, Fe atoms are oxidized and go from an **oxidation state** of +2 to +3 (ie, 1 electron lost per $\text{Fe}^{2+}(\text{aq})$).



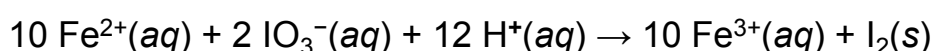
Simultaneously, iodine atoms are reduced and go from an **oxidation state of +5** to 0 (ie, 5 electrons gained per 1 iodine atom, or 10 electrons gained per 2 iodine atoms).



To cancel the electron terms in the overall reaction, multiply the oxidation half-reaction by 10 and then add the two half-reactions together.



Next, the **net charges** of the reactants and products **must be equal**. In the overall reaction, the net charge of the reactants is $10(+2) + 2(-1) = +18$ and the net charge of the products is $10(+3) + 1(0) = +30$. Because the reaction occurs in an aqueous solution under acidic conditions, $\text{H}^{+}(\text{aq})$ is added to balance the charges on the reactant side:



The final step is to balance the O atoms by adding $\text{H}_2\text{O}(\text{l})$ to the product side.

Therefore, the fully balanced redox reaction is:



(Choice A) In this reaction, the net charge is not balanced (ie, +12 for the reactants and +3 for the products).

(Choice B) The iodine atoms are not balanced in this equation. The reactant side has 1 iodine atom and the product side has 2 iodine atoms.

(Choice D) This is a balanced reaction under *basic conditions* (ie, adding $\text{OH}^{-}(\text{aq})$).

Educational objective:

An overall redox reaction can be expressed as the sum of the balanced oxidation and reduction half-reactions. When an overall redox reaction is fully balanced, the same number of each type of atom and the same net charge are present on both sides of the reaction equation.

Time spent: 0 sec

Subject:
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